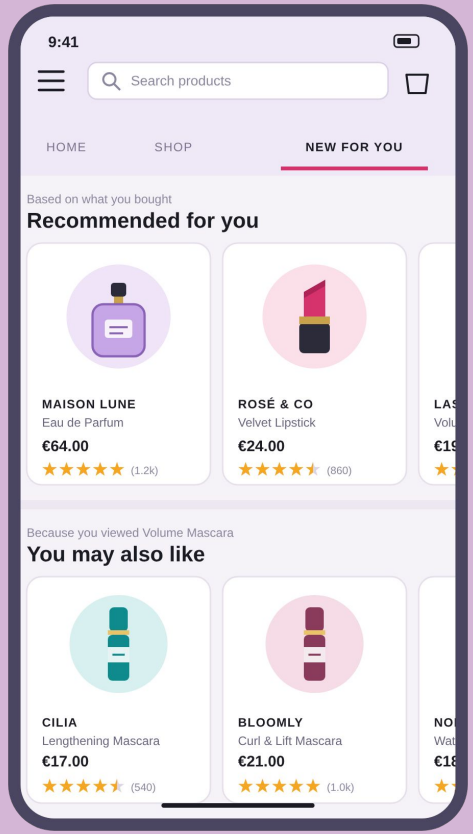
SEPHORA RECOMMENDER SYSTEM

Can a more complex model better predict what a customer will like?

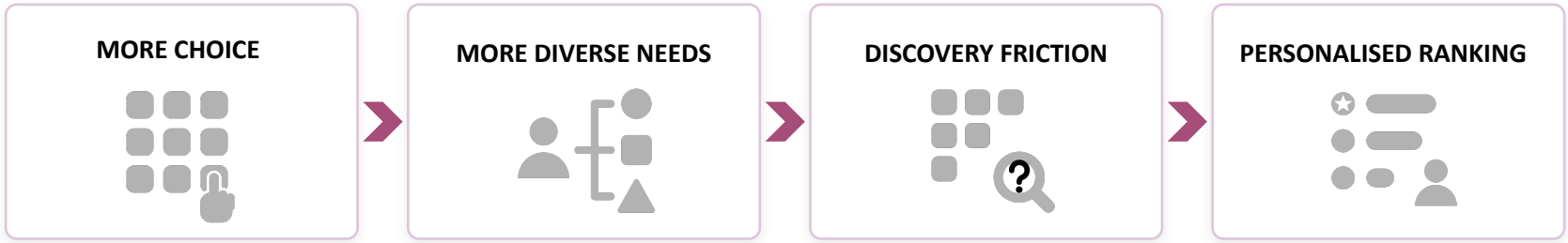
Comparative intelligence and ethics in machine learning systems



Machine Learning Module - October 2026

BUSINESS CONTEXT

Growing product choice demands more relevance



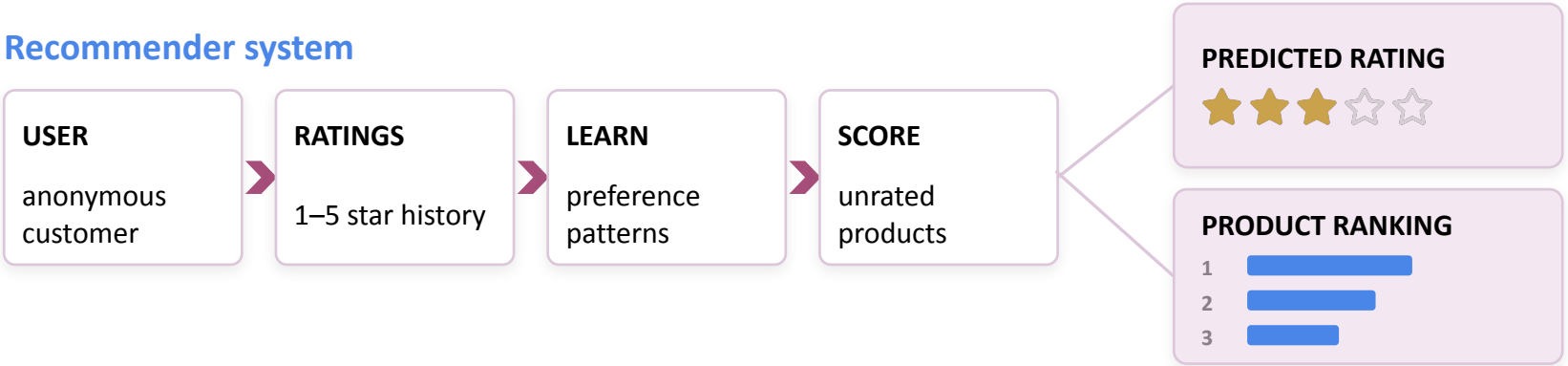
BUSINESS QUESTION

How can explicit customer rating behaviour support personalised product discovery across Sephora's product catalogue?

McKinsey & Company (2025); McKinsey & Company (2026)

ANALYTICAL GOAL

Rating history powers predictions and personalised ranking



MACHINE LEARNING QUESTION

How effectively can classical machine learning and deep learning predict customer rating and rank relevant Sephora products?

Jannach *et al.* (2010)

SEPHORA DATASET

Explicit ratings and product metadata fit both evaluation tasks

SOURCE (Inky, 2023)

Sephora Skincare Reviews - Kaggle
(scraped March 2023)

Licensed:
Creative Commons Attribution 4.0 International

18 ATTRIBUTES

Interactions: *user · product rating (1-5) · date*
Products: *name · brand · price*
Profile: *skin tone / type · eye / hair colour*
Reviews: *out of scope*

SCALE

1,1 million review records (2008-2023)

503,216 unique author IDs

2,351 unique product IDs

LIMITATION

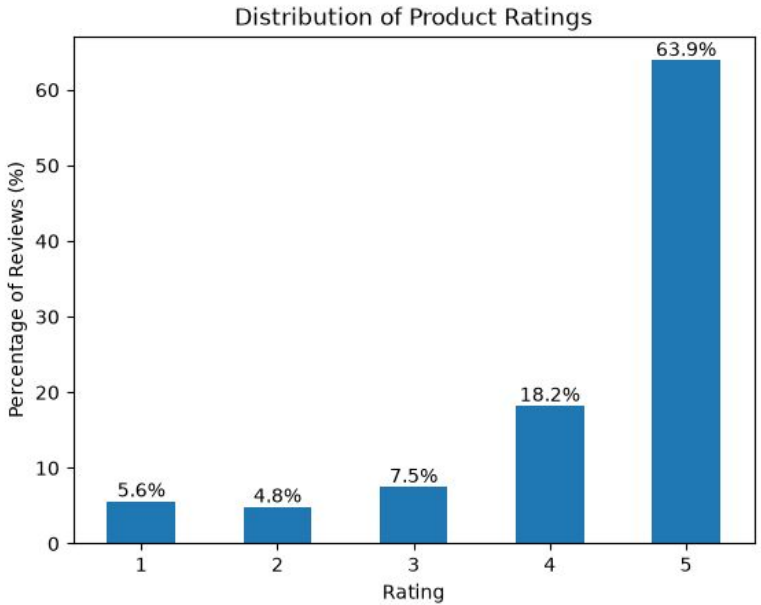
Self-selected explicit preference signals

Missingness on profile information *(e.g. skin tone)*

DATASET EXPLORATION

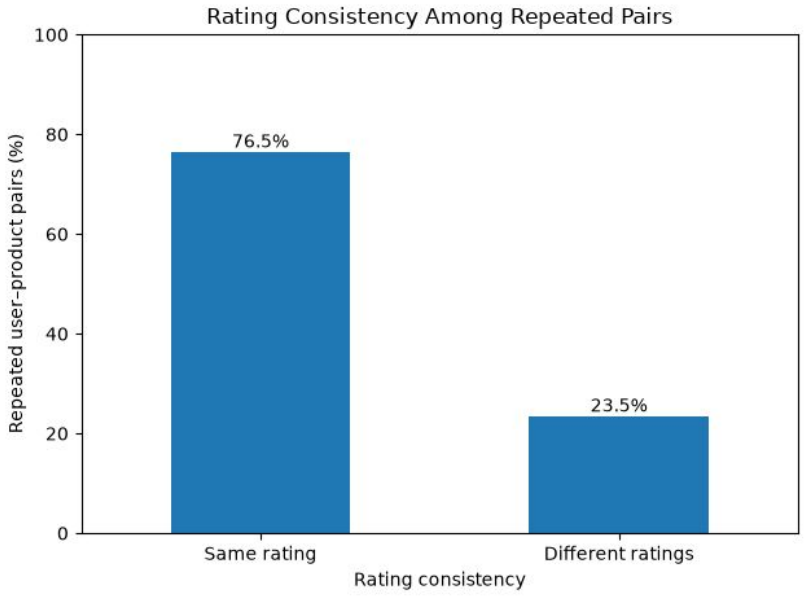
Ratings are mostly favourable, some preferences change over time

82% of 4–5 rating concentration



Source: Author's own work based on Inky (2023)

0.96% of repeated user-product interactions

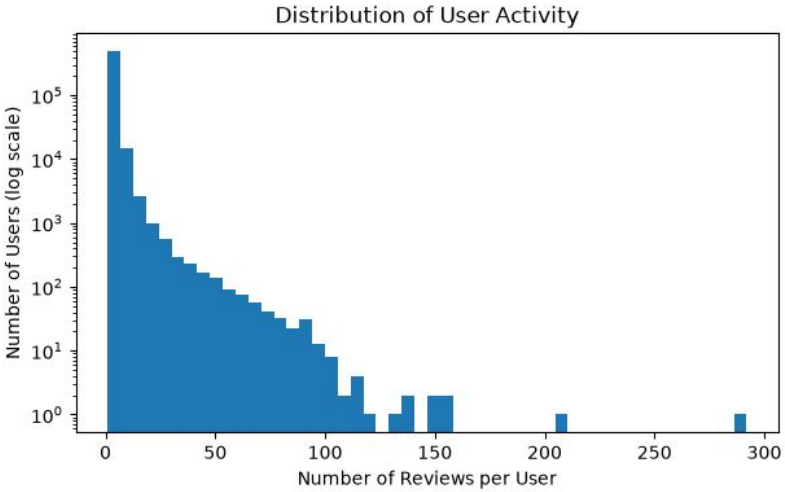


Source: Author's own work based on Inky (2023)

DATASET EXPLORATION

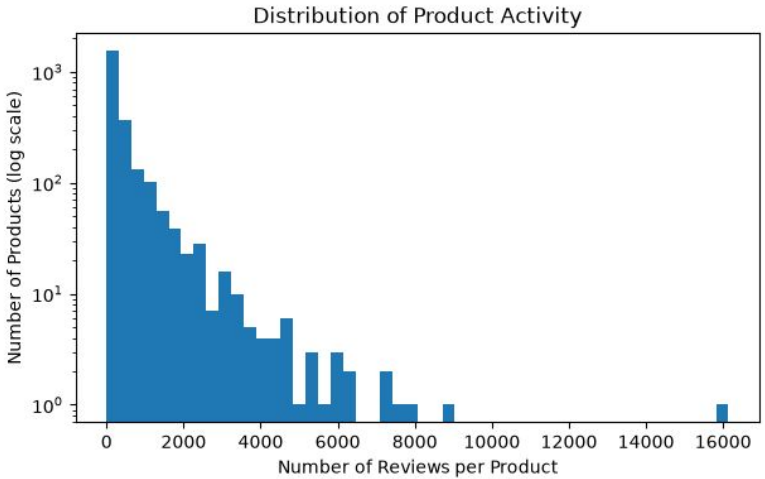
Sparse and uneven histories limit personalisation

58% of users leave only one review



Source: Author's own work based on Inky (2023)

Median **164**, maximum **>16K** reviewed products



Source: Author's own work based on Inky (2023)

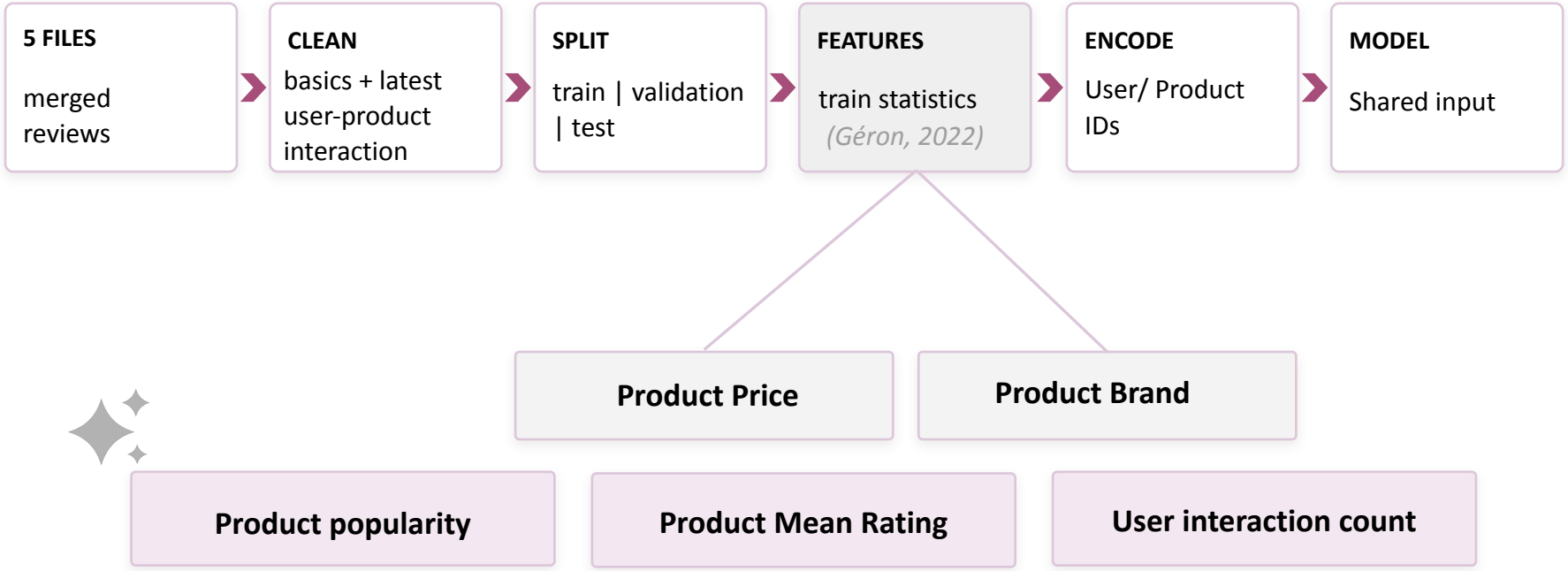
99.91% sparse user–product interactions

Review activities surged after 2018

DATA PREPARATION

Raw reviews become model-ready data

Preprocessing Pipeline

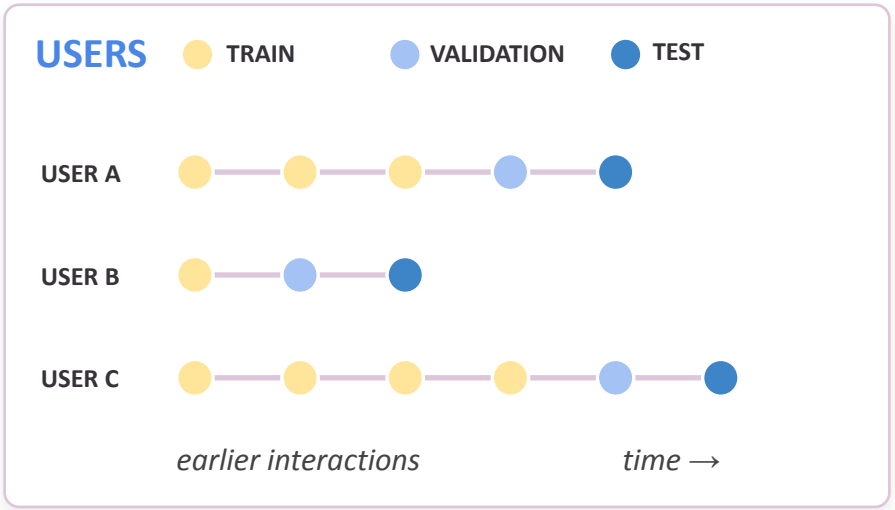


5 features reserved for feature-aware model

VALIDATION STRATEGY

Temporal leave-last-out preserves user's interaction order

Eligible users ≥ 3 interactions per user



PRODUCTS

Trained on: A B C D

Validated / tested on: B D

Product E:

Evaluation scope: Warm-start only, users and products without training history are not evaluated.

Cañamares, Castells and Moffat (2020); Meng *et al.* (2020); Yuan and Hernandez (2023)

MODEL SELECTION & DEVELOPMENT

Does greater complexity improve recommendations?

BENCHMARK	CLASSICAL MACHINE LEARNING	DEEP LEARNING	DEEP LEARNING EXTENSION
Global Mean Naive baseline No personalisation	**Biased Matrix Factorisation (Biased MF)** Embeddings + dot product + Biases Koren, Bell and Volinsky (2009)	**Neural Collaborative Filtering (Based NCF)** Embeddings + neural network He et al. (2017)	**Feature-aware NCF** Embeddings + 5 features + neural network Chen et al. (2020)
	CORE COMPARISON		
			EXTENSION

- **Implementation:** All models implemented in Pytorch for consistent comparison (Paszke *et al.*, 2019)
- **Training strategy:** Adam optimisation, validation-based hyperparameter selection, early stopping and regularisation
- **Initial configurations:** Informed by Singh, Dwivedi and Gautam (2026) and Zhang *et al.* (2023) tuned on validation data

PERFORMANCE EVALUATION

Two metrics answer two different questions

RMSE ↓

(Root Mean Square Error)

Rating Accuracy

How close is the predicted rating to the actual rating?

NDCG@10 ↑

(Normalised Discounted Cumulative Gain at rank 10)

Ranking quality

Does the relevant product appear near the top of the ranking?

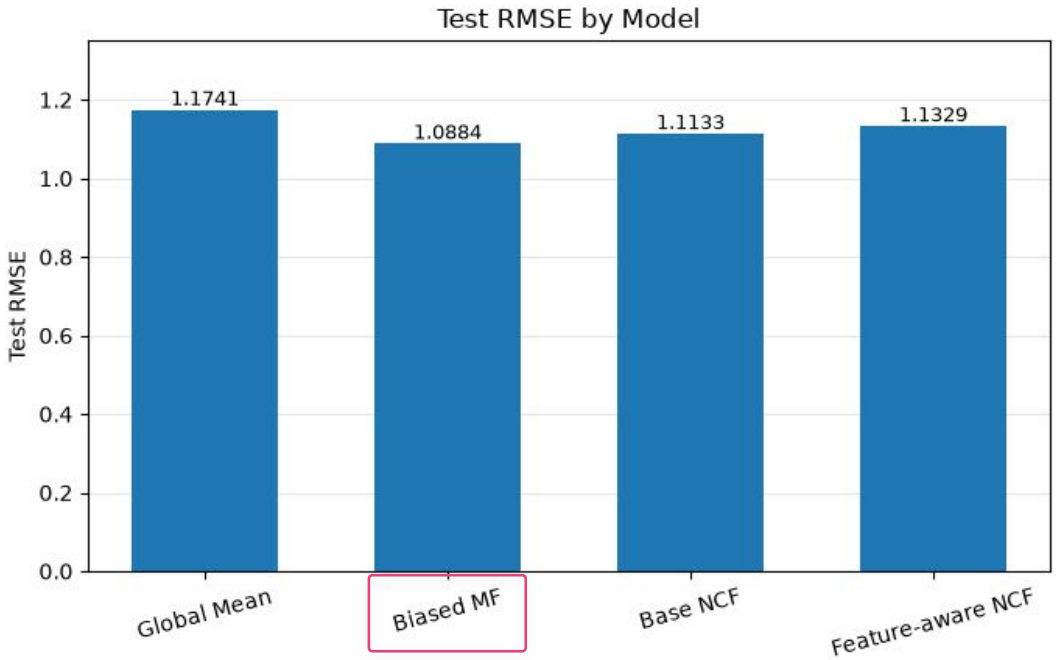
Sampled Protocol for NDCG@10 (Ghosh *et al.*, 2026)

1 test product *Rating >= 4* + **99 unseen products** → **Rank 100** → **Top 10 evaluation**

Unseen / Unrated product ≠ Disliked (He et al., 2017)

TEST RESULTS - RATING PREDICTION

Biased MF predicts ratings most accurately



Source: Author's own work based on Inky (2023)

BEST RMSE

Biased MF
1.0884

ACCURACY COMPARISON

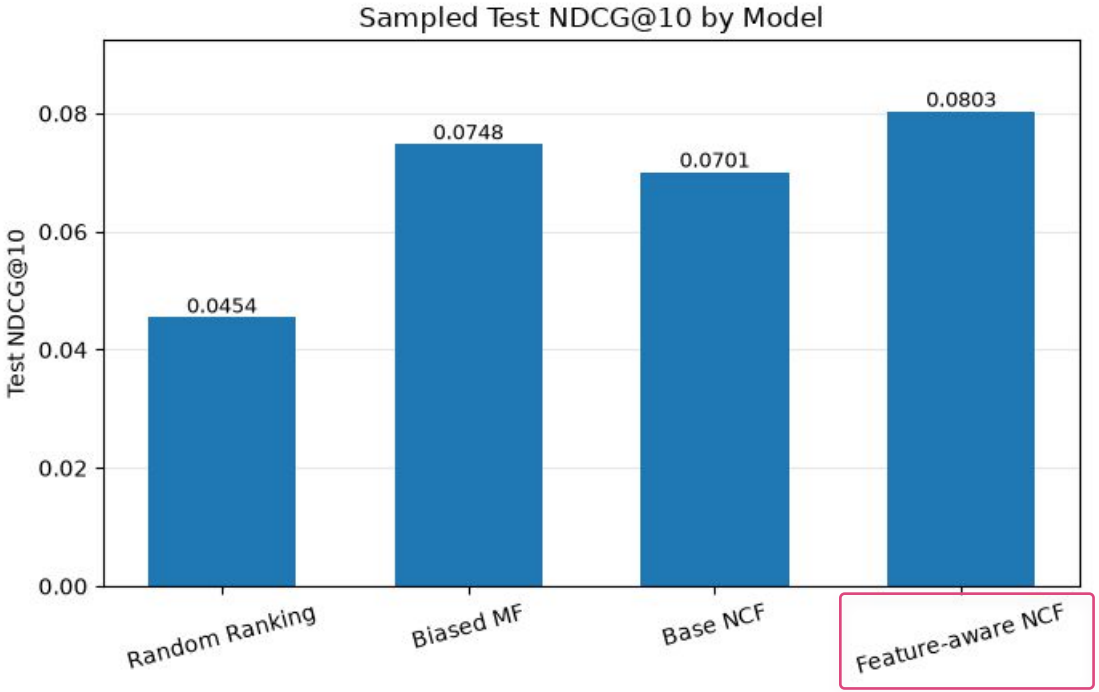
Neural complexity did not improve rating-prediction accuracy.

TRAINING BEHAVIOUR

Biased MF needs more training to run to achieve its best results as opposed to neural variants.

TEST RESULTS - RATING PREDICTION

Feature-aware NCF ranks the relevant product best



Source: Author's own work based on Inky (2023)

BEST NDCG@10

Feature-aware NCF
0.0803

RATING COMPARISON

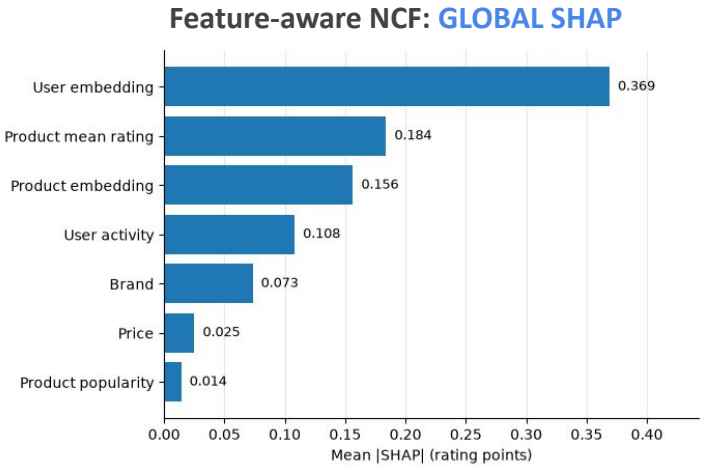
Feature-aware NCF ranks best ahead of Biased MF despite it's stronger rating accuracy.

COMPETITIVE INSIGHTS

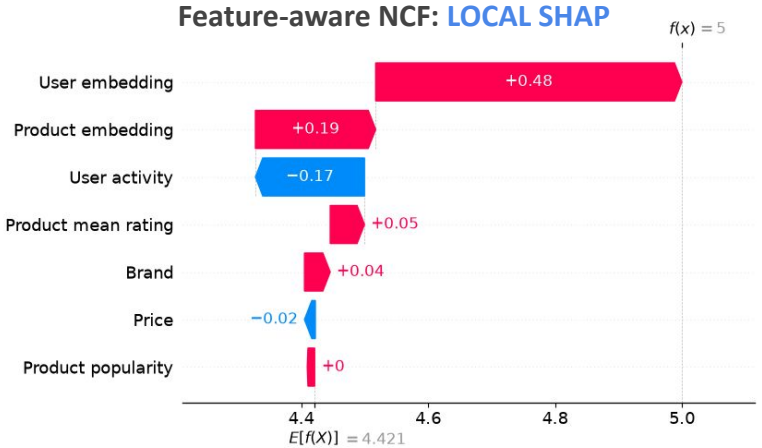
context supports relevant product discovery.

EXPLAINABILITY

User and product embeddings dominate rating predictions



Source: Author's own work based on Inky (2023)



Source: Author's own work based on Inky (2023)

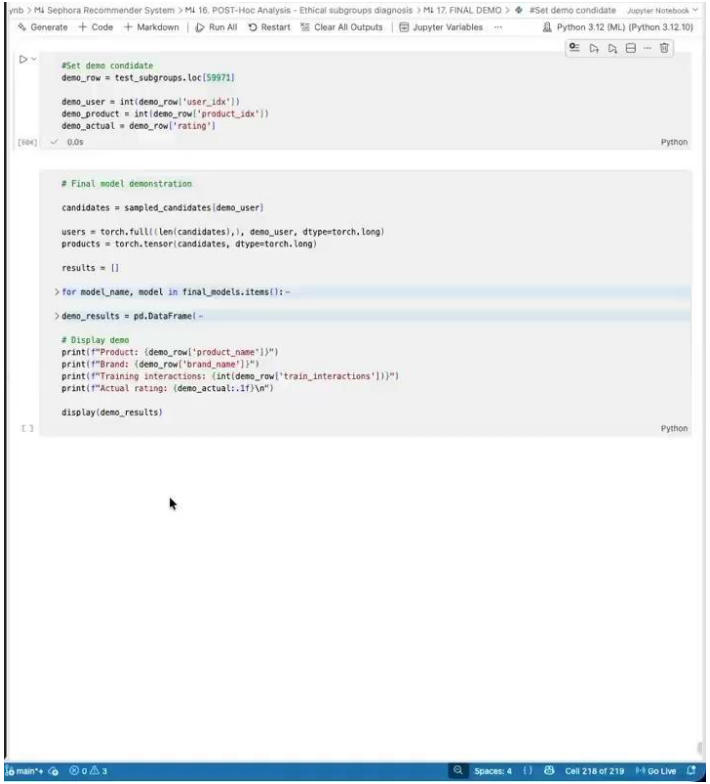
56.5%	43.5%
User + product embeddings	*Explicit + engineered features*

Actual rating: 5 | Predicted rating: 5.0 | Absolute error 0.0

Lundberg and Lee (2017); Molnar (2023); SHAP (n.d. -a, n.d. -b)

MODELS PERFORMANCE DEMONSTRATION

Biased MF wins on rating accuracy and rank in this example



Source: Author's own work based on Inky (2023)

EXAMPLE:

High activity user: 8 training interactions
Held-out product: 5 star rating
Candidate set: 100 products



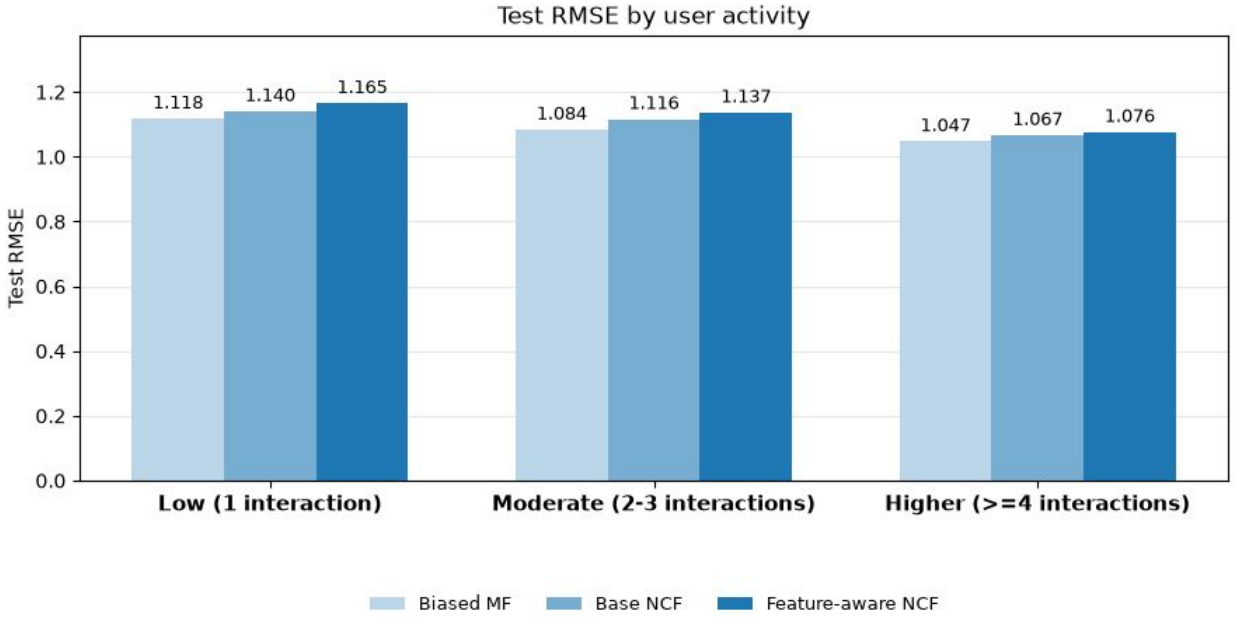
Image source:
Sephora France (n.d.).

Product: Honey Infused Lip Oil
Brand: Gisou

RESULTS	Rating	RANK
Biased MF	**4.61**	**7**
Base NCF	3.93	59
Feature-aware NCF	3.33	83

SUBGROUP ANALYSIS

Rating predictions are less accurate for low activity users



Source: Author's own work based on Inky (2023)

CONSISTENCY

Biased MF remains strongest on RMSE across groups

IMPLICATION

Sparse-history users are harder to serve

BIAS AND HARMS | DATA PRIVACY | FAIRNESS

Model performance alone does not ensure responsible use

BIAS + REPRESENTATION

- **Sparse history** users face higher prediction error
- **90% of skin-tone coverage, uneven representation** of skin tone group sizes
- **Cold-start product and users** with no training interaction are excluded

PRIVACY + PROVENANCE

- **Reviews are publicly scraped** and original-user consent is uncertain
- **Opaque author IDs are not proof of anonymity** (Narayanan and Shmatikov, 2008)
- **Self-reported attributes** may be incomplete or inaccurate

ALGORITHMIC FAIRNESS

- **Wrong product shown** monetary cost, bad experience and possible skin reaction (Lee, 2024)
- **Relevant product missed** less relevant catalogue and recommendations

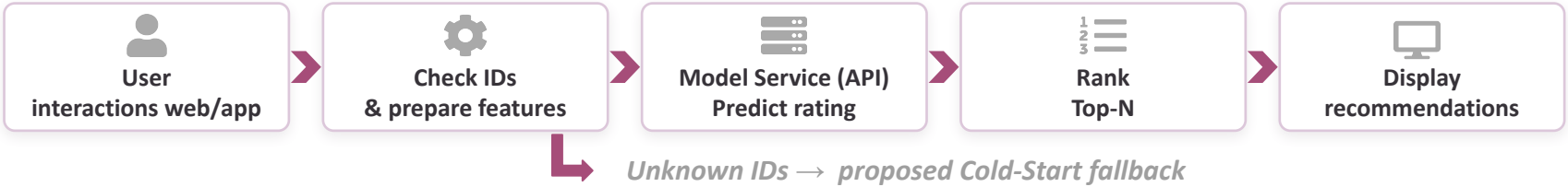
DEPLOYMENT CONSIDERATION

Feature-aware NCF is the preferred candidate for a monitored ranking pilot

FEATURE-AWARE NCF — Ranking pilot candidate

BIASED MF — Evaluation baseline only

Proposed Deployment



Monitor and Retrain

Monitor Ranking quality | Subgroup rating accuracy | Fallback usage | Popularity-bias drift | Recommendation popularity

Retrain | **Validate before release**

Géron (2022); Kreuzberger, Kühl and Hirschl (2023)

References

REFERENCES

Cañamares, R., Castells, P. and Moffat, A. (2020) ‘Offline evaluation options for recommender systems’, *Information Retrieval Journal*, 23(4), pp. 387–410. Available at: https://doi.org/10.1007/s10791-020-09371-3

Chen, W.-H., *et al.* (2020) 'Attribute-Aware Recommender System Based on Collaborative Filtering: Survey and Classification', *Frontiers in Big Data*, 2, Article 49. Available at: https://doi.org/10.3389/fdata.2019.00049

Géron, A. (2022) *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd edn. Sebastopol, CA: O'Reilly Media.

Ghosh, S. *et al.* (2026) ‘Dual stream adaptive sequential recommendation’, *Discover Artificial Intelligence*, 6, article 1201. Available at: https://doi.org/10.1007/s44163-026-02282-0

Harris, C.R. *et al.* (2020) ‘Array programming with NumPy’, *Nature*, 585, pp. 357–362. Available at: https://doi.org/10.1038/s41586-020-2649-2

He, X., *et al.* (2017) ‘Neural collaborative filtering’, *Proceedings of the 26th International Conference on World Wide Web (WWW ’17)*, Perth, Australia, pp. 173–182. Available at: https://doi.org/10.1145/3038912.3052569

Inky, N. (2023) *Sephora products and skincare reviews* [dataset]. Kaggle. Available at: https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews (Accessed 13 September 2026).

Jannach, D., *et al.* (2010) *Recommender systems: An introduction*. Cambridge: Cambridge University Press. Available at: https://doi.org/10.1017/CBO9780511763113

Koren, Y., Bell, R. and Volinsky, C. (2009) ‘Matrix factorization techniques for recommender systems’, *Computer*, 42(8), pp. 42–49. Available at: https://doi.org/10.1109/MC.2009.263

Kreuzberger, D., Kühl, N. and Hirschl, S. (2023) ‘Machine learning operations (MLOps): Overview, definition, and architecture’, *IEEE Access*, 11, pp. 31866–31879. Available at: https://doi.org/10.1109/ACCESS.2023.3262138

REFERENCES

Lee, J., *et al.* (2024) 'Deep learning-based skin care product recommendation: A focus on cosmetic ingredient analysis and facial skin conditions', *Journal of Cosmetic Dermatology*, 23(6), pp. 2066–2077. Available at: https://doi.org/10.1111/jocd.16218

Lundberg, S.M. and Lee, S.-I. (2017) 'A unified approach to interpreting model predictions', *Advances in Neural Information Processing Systems*, 30. Available at: https://arxiv.org/abs/1705.07874 (Accessed: 3 October 2026).

McKinsey & Company (2026) *From aisle to algorithm: The beauty categories, channels, and concepts shaping 2030 growth*. 18 June. Available at: https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/state-of-beauty-2025 (Accessed: 3 October 2026).

McKinsey & Company (2025) *A close look at the global beauty industry in 2025*. 28 August. Available at: https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/a-close-look-at-the-global-beauty-industry-in-2025 (Accessed: 3 October 2026).

Meng, Z., *et al.* (2020) 'Exploring data splitting strategies for the evaluation of recommendation models', *arXiv* [Preprint]. Available at: https://doi.org/10.48550/arXiv.2007.13237

Molnar, C. (2023) *Interpretable machine learning: A guide for making black box models explainable*. 2nd edn. Available at: https://christophm.github.io/interpretable-ml-book (Accessed: 3 October 2026).

Narayanan, A. and Shmatikov, V. (2008) 'Robust de-anonymization of large sparse datasets', *2008 IEEE Symposium on Security and Privacy (SP 2008)*, Oakland, CA, pp. 111–125. Available at: https://doi.org/10.1109/SP.2008.33

Pedregosa, F. *et al.* (2011) 'Scikit-learn: Machine learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830. Available at: https://www.jmlr.org/papers/v12/pedregosa11a.html (Accessed: 3 October 2026).

Sephora France (n.d.) Honey Infused Lip Oil – Huile lèvres au miel [Product image]. Available at: https://www.sephora.fr/p/honey-infused-lip-oil---huile-levres-au-miel-720759.html (Accessed: 7 October 2026).

REFERENCES

SHAP (n.d.-a) *API examples: Plots*. Available at: https://shap.readthedocs.io/en/latest/api_examples.html#plots (Accessed: 2 October 2026).

SHAP (n.d.-b) 'shap.DeepExplainer', *SHAP documentation*. Available at: https://shap.readthedocs.io/en/latest/generated/shap.DeepExplainer.html (Accessed: 2 October 2026).

Singh, A.P., Dwivedi, V.K. and Gautam, C.S. (2026) 'Neural collaborative filtering-based personalized recommendation system using Amazon Beauty dataset', *International Research Journal of Computer Science*, 13(6), pp. 592–600. Available at: https://www.irjcs.com/volumes/Vol13/iss-06/01.JNCS10083.pdf (Accessed: 3 October 2026).

Yuan, H. and Hernandez, A.A. (2023) 'User cold start problem in recommendation systems: A systematic review', IEEE Access, 11, pp. 136958–136977. Available at: https://doi.org/10.1109/ACCESS.2023.3338705.

Zhang, A., *et al. (*2023) *Dive into deep learning*. Cambridge: Cambridge University Press. Section 21.3, 'Matrix factorization'. Available at: https://d2l.ai/chapter_recommender-systems/mf.html (Accessed: 3 October 2026).

Appendix

Original review attributes in the modelling scope and their role in the recommender project

Attributes	Description	Recommender Relevance
author_id	Unique Reviewer Identifier	Core interactions, model input, encoded as ID
product_id	Unique Product Identifier	Core interactions, model input, encoded as ID
rating	1-5-star rating	Prediction Target and evaluation
submission_time	Review timestamp	Chronological splitting and last interaction
product_name	Product Name	Product interpretation only
brand_name	Product Brand	Brand embedding, Additional feature
price_usd	Product Price	Numerical feature
skin_tone	Self-reported skin tone	Subgroup analysis only, not model input, 16% missingness

Source: Inky (2023)

Original review attributes outside of the modelling scope

Attributes	Description	Exclusion Reason
is_recommended	Explicit recommendation indicator	Closely related to the rating target
helpfulness	Review helpfulness measure	Review engagement outside of modelling scope
total_feedback_count	Total feedback received for a review	Review engagement outside of modelling scope
total_neg_feedback_count	Negative feedback received by a review	Review engagement outside of modelling scope
total_pos_feedback_count	Positive feedback received by a review	Review engagement outside of modelling scope
review_text	Review content	Text modelling, outside of modelling scope
review_title	Review title	Text modelling, outside of modelling scope
eye_color	Self-reported eye colour	Weaker feature, outside of selected feature scope
hair_color	Self-reported hair colour	Weaker feature, outside of selected feature scope
skin_type	Self-reported skin type	Initially considered but not used for subgroup analysis due to relevance and missingness (10%

Source: Inky (2023)

MODEL SELECTION AND COMPARISON *(literature-informed)*

Dimensions	Biased Matrix Factorisation (Biased MF)	Neural Collaborative Filtering (Base NCF)	Feature-aware NCF
Input Data	User–item ratings	User–item interactions	Interactions + user/item attributes
Interaction modelling	Linear dot product with biases	Nonlinear neural interaction	Neural interaction + attributes
Sparse data	Limited interactions remains a challenge	Neural complexity does not remove sparsity	Attributes may support limited interaction evidence
Rating tendencies	Explicit user and item biases	No explicit biases	Attributes may capture additional rating patterns
Capacity	Controlled by latent dimensions	Controlled by embeddings and network size	Depends on attributes representations
Interpretability	Bias terms/ latent factors inspectable	Learned representations difficult to interpret, requires additional tools e.g. SHAP	Attributes offer context, but do not provide transparency, additional tools are required (e.g. SHAP)

Source: Chen *et al.* (2020); He *et al.* (2017); Koren, Bell and Volinsky (2009); Zhang *et al.* (2023)

MODELS HYPERPARAMETERS TUNING

Model	Initial configuration	Candidate tested	Selected Configuration	Best Validation RMSE + Epochs
Biased MF	Latent factors 30 Learning rate: 0.002 Weight decay: 1e-05 Max epochs: 30 Early stopping: 3	Latent factors 10, 30, 50, 70, 90 Learning rate: 0.001, 0.002, 0.005 Weight decay: 1e-06, 1e-05, 1e-04 Max epochs: 50 *(limit increased from 30 to 50 during further tuning)* Early stopping: 3	Latent factors 90 Learning rate: 0.002 Weight decay: 1e-06 Max epochs: 50 Early stopping: 3	RMSE: 0.9726 Best epoch: 43
Base NCF	Embedding dimension: 64 Hidden layer: (128, 64) Learning rate: 0.002 Weight decay: 1e-05 Dropout: 0 Max epochs: 30 Early stopping: 3	Embedding dimension: 32, 64, 96 Hidden layer: Fixed Learning rate: 0.001, 0.002, 0.005 Weight decay: 1e-06, 1e-05, 1e-04 Dropout: 0.0, 0.1, 0.2, 0.3 Max epochs: 30 Early stopping: 3	Embedding dimension: 96 Hidden layer: (128, 64) Learning rate: 0.002 Weight decay: 1e-04 Dropout: 0.0 Max epochs: 30 Early stopping: 3	RMSE: 1.0328 Best epoch: 2
Feature-aware NCF	Started from selected Base NCF settings and tuned separately after early overfitting.	Embedding dimension: Fixed Hidden layer: Fixed Learning rate: 0.0002 and 0.0005, 0.001, 0.002 Weight decay: 1e-04, 5e-4, 1e-03 Dropout: 0.0, 0.1, 0.2, 0.3 Max epochs: 30 Early stopping: 3	Embedding dimension: 96 Hidden layer: (128, 64) Learning rate: 0.0002 Weight decay: 1e-03 Dropout: 0.0 Max epochs: 30 Early stopping: 3	RMSE: 1.0480 Best epoch: 3

Initial configuration informed by Singh, Dwivedi and Gautam (2026) and Zhang *et al.* (2023)

SKIN TONE ANALYSIS (TEST RMSE based)

Model	Skin tone	Count	Biased MF	Base NCF	Feature-aware NCF
0	dark	46	1.2711	1.2828	1.2720
1	deep	1959	1.1016	1.1227	1.1519
2	fair	21659	1.1026	1.1264	1.1486
3	fairLight	5879	1.1198	1.1465	1.1552
4	light	27858	1.1031	1.1282	1.1460
5	lightMedium	19033	1.0801	1.1062	1.1212
6	medium	7024	1.0686	1.0955	1.1157
7	mediumTan	5992	1.1297	1.1507	1.1706
8	notSureST	7	1.1243	1.1584	1.0591
9	olive	131	1.3567	1.3482	1.3316
10	porcelain	160	1.2972	1.2356	1.2771
11	rich	487	0.9989	1.0557	1.0634
12	tan	3228	1.0327	1.0735	1.1028

lowest RMSE highest RMSE

Skin tone is self-reported, with 90% of coverage in test observations.

12 skin tone groups are unevenly represented, with no clear performance pattern.

These differences do not alone establish fairness or unfairness and monitoring across the groups is required.

TOOLS AND LIBRARIES USED

Tool / Library	Role and Justification
PyTorch	Flexible embeddings and neural layers supported all three models within a shared training function, enabling a consistent comparison (Paszke *et al*., 2019).
scikit-learn	Provided root_mean_squared_error for the global-mean baseline and subgroup evaluation (Pedregosa *et al.,* 2011).
SHAP	DeepExplainer provided approximate SHAP attributions to explain the Feature-aware NCF model's predicted ratings (Lundberg and Lee, 2017, SHAP (n.d.-b).
pandas	Supported data loading, cleaning, grouping, dataset splitting and results tables.
NumPy	Provided numerical operations and seeded sampling of unrated candidate products for sampled NDCG evaluation (Harris *et al.*, 2020).
Matplotlib / Seaborn	Produced EDA, model-comparison and diagnostic charts.